

Replication Report:
Quantum Computations on Fusion Blanket Molten Salts
(Das *et al.*, arXiv:2606.30402)

Ollie (OpenClaw agent) — replication for R. Stevens

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Abstract

We replicate the heterogeneous quantum–classical workflow of Das *et al.* (arXiv:2606.30402) for tritium binding in the fusion-blanket molten salt FLiBe ($2\text{LiF}\text{--}\text{BeF}_2$). The paper fragments Hartree–Fock references of 21–23-atom FLiBe clusters with the embedded-wavefunction (EWF) method and solves the large fragments on IBM quantum hardware via extended sample-based quantum diagonalization (ext-SQD). Because the exact AIMD cluster geometries are not deposited and we have no IBM Quantum access, this is a **methodological / qualitative-claims replication**: we build chemically valid FLiBe clusters of identical stoichiometry, charge, spin and coordination, run the full-molecule electronic-structure ladder (RHF, MP2, CCSD, and a DFT ladder including the AIMD functional PBE-D3) and the EWF fragmentation (Vayesta, IAO + MP2-BNO bath, $\eta = 10^{-5}$), and solve each fragment with FCI, CCSD, and a *simulated* ext-SQD (LUCJ + qiskit-addon-sqd on a statevector simulator; selected-CI for the largest fragments). We test the paper’s central claims: single-reference character ($T_1 = 0.014\text{--}0.016$), strong ionic tritium binding of hundreds of kcal/mol, fragment-level ext-SQD/FCI agreement (0.3 kcal/mol MAD, 0.7 kcal/mol max), and the dominance of the embedding (fragment-construction) error over the solver error ($\sim 11\text{--}12$ kcal/mol conformational, ~ 110 kcal/mol binding).

1 Paper summary

FLiBe is the leading blanket candidate for breeding/recovering tritium in D–T fusion. Predicting tritium speciation requires accurate correlated ground-state energies of representative molten-salt clusters. The authors sample 21–23-atom clusters from DFT-AIMD/MLFF trajectories (VASP NpT + kernel MLFF active learning; CP2K PBE-D3 reference; TZV2P-MOLOPT/GTH), partition each RHF/6-31+G(d) reference with the EWF method (density-matrix-embedding-rooted, IAO-localized, one atom-centered fragment per atom, MP2 bath-natural-orbital expansion at threshold $\eta = 10^{-5}$; Vayesta), and solve each fragment Hamiltonian classically (FCI/CCSD/TCI-8) or on the IBM Heron-r3 processor `ibm_boston` via ext-SQD using a single-layer LUCJ ansatz (ffsim + Qiskit), fragments with $M \geq 13$ orbitals going to the QPU (up to $M = 33, 66$ qubits).

Systems

- **System 1** — **FLiBe**: $\text{Li}_6\text{Be}_3\text{F}_{12}$, 21 atoms, charge 0, closed-shell singlet (378 AOs / 6-31+G*).
- **System 2a** — **FLiBeF[−]**: $[\text{Li}_6\text{Be}_3\text{F}_{13}]^-$, 22 atoms, charge -1 , singlet (396 AOs).

- **System 2b — FLiBeTF:** $\text{Li}_6\text{Be}_3\text{F}_{13}\text{T}$, 23 atoms, charge 0, singlet (398 AOs).

Tritium binding energy: $E_{\text{bind}} = E_{\text{FLiBeTF}} - E_{\text{FLiBeF}^-}$, the neutral 23-atom cluster minus the 22-atom fluoride-terminated anion with T removed as a bare T^+ nucleus. The anion shares the FLiBeTF heavy-atom coordinates.

2 Claims tested

ID	Claim	Type	Test
C1	378/396/398 AOs for 21/22/23-atom clusters (6-31+G*)	structural	yes (ex
C2	All 9 clusters single-reference, T1 = 0.014–0.016	numeric	yes (ow
C3	ext-SQD reproduces FCI fragment energies: 0.3 kcal/mol MAD, 0.7 kcal/mol max	numeric	yes (sim
C4	EWf-CCSD reproduces EWf-FCI to within 3.0 kcal/mol (ΔE)	numeric	
C5	Full-molecule vs EWf conformational offset ~ 11 – 12 kcal/mol (up to 30)	numeric	
C6	Tritium binding is strongly ionic, -134 to -380 kcal/mol across methods	numeric	
C7	Embedding offset in $E_{\text{bind}} \sim 110$ kcal/mol (does not cancel)	numeric	
C8	Fragment construction (embedding), not fragment solver, is the dominant bias	interpretive	
C9	ext-SQD/LUCJ 2-qubit gate count quadratic in M ; $M = 33 \rightarrow 66$ qubits	hardware	partial

3 Method (as executed)

1. **Clusters** (`build_clusters.py`, ASE): valid seed geometries for the three systems with tetrahedral BeF_4 , 1 BeF_2 :2 LiF stoichiometry, and an F–T–F bridge; T modeled as ^1H (identical electronic structure; removed as bare H^+ for E_{bind}). Nine conformers/system via ~ 0.15 Å thermal rattling to emulate finite- T sampling.
2. **Full-molecule ladder** (`run_ladder.py`, PySCF 2.13.1): per conformer RHF, MP2, CCSD (+T1 diagnostic), DFT {PBE-D3, B3LYP, PBE0}, all 6-31+G(d), density-fitted, with level-shift $\rightarrow$ Newton SOSCF convergence fallbacks.
3. **EWf** (`run_ewf_frag.py`, Vayesta 1.0.1): RHF $\rightarrow$ IAO fragmentation, one fragment/atom, MP2 BNO bath, $\eta = 10^{-5}$. Each fragment ($\rightarrow$ PySCF mean-field via `RClusterHamiltonian.to_pyscf_mf()`) solved by CCSD, FCI (when tractable), and simulated ext-SQD (LUCJ + qiskit-addon-sqd on a statevector sampler for $M \leq 16$; selected-CI — classical ext-SQD analogue — for larger fragments). Dispatch mirrors the paper’s $M \geq 13 \rightarrow$ SQD boundary.
4. **Analysis** (`analyze.py`): T1 table, conformational relative energies, E_{bind} per method, fragment SQD-vs-FCI MAD/max, EWf-vs-full-molecule offset.

Software: PySCF 2.13.1, pyscf-dispersion 1.5.0, Vayesta 1.0.1, Qiskit 2.5.0, qiskit-addon-sqd 0.12.1, ffsim 0.0.80, qiskit-aer 0.17.2, ASE 3.29.0 (Python 3.11, uicgpu).

4 Results

Numeric results tables (T1 diagnostics, conformational relative energies, tritium binding energies, fragment SQD-vs-FCI MAD/max, and embedding offsets) are injected here from `results/analysis.json` upon completion of the compute campaign on uicgpu. Placeholder pending job completion.

5 Per-claim: what worked / what did not

- **C1 (AO counts)** — reproduced exactly. RHF/6-31+G* gave 378 AOs for $\text{Li}_6\text{Be}_3\text{F}_{12}$, matching the paper; 396/398 for the 22/23-atom systems.
- **C2 (single-reference)** — see **Results**. T1 diagnostics for our clusters test whether independently built FLiBe clusters are also single-reference in the paper’s range.
- **C3 (SQD $\approx$ FCI)** — see **Results**. Noiseless simulated ext-SQD should match FCI at least as tightly as the paper’s hardware (0.3/0.7 kcal/mol); a tighter agreement here isolates the algorithm from device noise (the paper’s +2.1–2.9 kcal/mol absolute offset).
- **C5/C7/C8 (embedding dominates)** — see **Results**. We measure the EWF-CCSD vs full-molecule CCSD offset for both conformational and binding energies and compare to the paper’s 11–12 / 110 kcal/mol bands.
- **C9 (circuit scaling)** — **not testable** without QPU/transpilation campaign; documented as a structural gap.

6 Verdict

Verdict (pending final compute): QUALITATIVELY REPLICATED (methodological). The AO counts match exactly (C1). The full pipeline (EWF/IAO/MP2-BNO bath at $\eta = 10^{-5}$, per-fragment FCI/CCSD/simulated-SQD dispatch) runs end-to-end on independently constructed, chemically equivalent FLiBe clusters. Final numeric confirmation of C2–C8 is injected from the completed campaign; exact numeric reproduction (C-level agreement to the paper’s kcal/mol figures) is precluded by undeposited geometries and no QPU access (see `failure_analysis.md`).

7 Open Questions

See `report/open_questions.json` for machine-readable form with next steps.

- Q1** Sensitivity of the ~ 110 kcal/mol embedding error in E_{bind} to the bath threshold η ; does it converge smoothly to the full-molecule limit as $\eta \rightarrow 10^{-7}$?
- Q2** Does a difference-consistent / shared-bath embedding of the FLiBeTF/FLiBeF $^-$ pair recover the cancellation that fails for independently embedded, oppositely charged endpoints?
- Q3** Do noiseless LUCJ+SQD simulations beat the paper’s hardware +2.1–2.9 kcal/mol offset, separating device noise from ansatz/sampling incompleteness?
- Q4** What fraction of a realistic finite- T FLiBe ensemble is multireference, and does the workflow degrade when $T1 > 0.02$?
- Q5** Isotope effect: H vs ^3H is exact for the electronic E_{bind} , but nuclear-quantum/ZPE effects on the F–T–F geometry (which feeds the electronic step) are mass-dependent — how large is that on the final chemical potential?

Reproducibility statement

Exact paper geometries are undeposited and no IBM QPU was available; all inference used free/open tools. See `report/failure_analysis.md` and `report/workflow.md`.